Product Line Profitability and Margin Performance Analysis for a Multi-Division Candy Distributor: A Power BI Business Intelligence Approach

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***Abstract*— For multi-division distributors, aggregate revenue and margin figures often conceal significant structural concentration risk. This paper presents an exploratory data analysis and interactive Power BI business intelligence dashboard developed for Nassau Candy Distributor to evaluate product-, division-, and geography-level profitability. Using 10,194 transaction records spanning January 2024 through December 2025, we construct a star-schema data model, define custom profitability measures (gross margin, profit contribution, cumulative Pareto share), and build a five-page interactive dashboard with dynamic threshold-based repricing flags. Results show that a single division (Chocolate) accounts for approximately 93% of revenue and 95% of gross profit, that five of fifteen products (33% of the catalog) drive approximately 80% of both profit and revenue, and that a single product (Kazookles) exhibits a critically low 7.69% gross margin against a 92.31% cost-to-sales ratio. Geographic revenue is similarly concentrated, with five states contributing 51% of total sales. We conclude with concrete recommendations for portfolio diversification, targeted repricing, and geographic risk mitigation.**

Index Terms*— *business intelligence, Power BI, profitability analysis, Pareto analysis, gross margin, data visualization, DAX, exploratory data analysis.

I. INTRODUCTION

Sales volume alone is a misleading indicator of financial health for a product distributor. A product may sell in high volume while generating disproportionately low profit, consume excessive manufacturing or operational cost, or otherwise appear successful on the surface while quietly eroding overall margin. Prior to this analysis, Nassau Candy Distributor lacked systematic visibility into which product lines deliver the highest gross margin, whether high-sales products are genuinely profitable, how profitability varies across product divisions, and which products represent an active margin risk. In the absence of this visibility, decisions regarding pricing, promotion, and product-portfolio rationalization have historically been reactive and intuition-driven rather than data-driven.

This paper documents the exploratory data analysis (EDA), data modeling methodology, and key findings underlying an interactive Power BI dashboard constructed to address this gap. Section II describes the dataset and cleaning methodology. Section III presents the analytical approach and key performance indicators. Section IV reports findings across division, product, cost, and geographic dimensions. Section V describes the dashboard implementation. Section VI presents recommendations, and Section VII concludes.

II. DATA AND METHODOLOGY

A. Dataset Description

The source dataset comprises 10,194 order-line transaction records spanning January 1, 2024 through December 31, 2025. Each record includes Order Date, Ship Date, Division, Region, State/Province, Product Name, Sales, Units, Cost, and Gross Profit. Three product divisions are represented (Chocolate, Sugar, Other) across fifteen distinct products and four sales regions (Atlantic, Gulf, Interior, Pacific).

B. Data Cleaning and Validation

Data preparation was performed in Power Query prior to modeling. Order Date and Ship Date fields were stored as DD-MM-YYYY text strings and required explicit locale-based parsing (English, United Kingdom) to avoid day/month transposition errors. Categorical text fields (Product Name, Division, Region, City, State/Province) were trimmed to remove stray whitespace that would otherwise fragment groupings in downstream visuals. The Geography dimension table required deduplication keyed specifically on Postal Code, rather than across all columns jointly, to satisfy the uniqueness constraint required for a valid one-to-many relationship with the fact table.

A dedicated Calendar dimension table was constructed to span the confirmed full range of Order Date. An initial single-year assumption was found, upon full-column profiling, to omit a substantial fraction of records and was corrected to span both 2024 and 2025. The Month column additionally required an explicit sort-by-column binding to a numeric MonthNo field; a secondary defect was traced to MonthNo being stored as a text type rather than a whole number, which produced lexicographic rather than chronological ordering (e.g., 1, 10, 11, 12, 2, 3) until corrected.

C. Data Quality Finding — Ship Date Anomaly

A systemic data quality issue was identified in the Ship Date field. While Order Date values fall within the expected 2024–2025 range, Ship Date values were found to span June 2026 through June 2030, in many cases more than five years after the corresponding order date. This offset affects the majority of records rather than a small isolated subset, indicating a data-generation or export artifact rather than genuine fulfillment delay. Because the present analysis is scoped to profitability and margin performance rather than logistics, Ship Date was excluded from all time-based relationships and visuals. This exclusion has no bearing on any profitability metric reported in this paper, as Sales, Cost, and Gross Profit are computed independently of shipping timing.

D. Data Model

A star-schema data model was constructed comprising one fact table (Orders) and three dimension tables (Products, Geography, Calendar), related as one-to-many with the Calendar table marked as the model's official date table. A What-If numeric parameter (Margin Threshold) was implemented to drive an interactive repricing-flag mechanism, described further in Section V.

E. Key Performance Indicators

TABLE I

CORE PROFITABILITY METRICS USED THROUGHOUT THIS ANALYSIS

Metric	Definition
Gross Margin %	Gross Profit ÷ Sales
Profit per Unit	Gross Profit ÷ Units
Revenue Contrib. %	Sales ÷ Total Sales
Cost-to-Sales Ratio	Cost ÷ Sales

III. ANALYTICAL APPROACH

Four complementary analyses were conducted: (1) product-level ranking by gross profit and margin to distinguish high-margin from high-volume/low-margin products; (2) division-level aggregation comparing revenue share against profit share to detect imbalance; (3) Pareto (80/20) concentration analysis on both cumulative profit and cumulative revenue, ranked by product; and (4) cost-structure diagnostics using a joint margin-threshold and cost-to-sales-ratio rule to flag products for repricing review. A DAX RANKX-based ranking measure combined with a cumulative-sum pattern was used to compute Pareto shares, with the underlying visual explicitly sorted by rank to ensure a monotonically increasing cumulative curve.

IV. FINDINGS

A. Division-Level Concentration

Despite the business operating three nominal divisions, Chocolate overwhelmingly dominates both revenue and profit, as summarized in Table II.

TABLE II

SALES AND PROFIT DISTRIBUTION BY DIVISION (% OF TOTAL)

Division	Sales %	Profit %
Chocolate	92.9%	95.1%
Other	6.8%	4.6%
Sugar	0.3%	0.3%

Margin percentage, in contrast, is comparatively balanced across divisions: Chocolate carries a 67.4% average margin, Sugar 66.6%, and Other 44.8%. Sugar and Other are therefore not inherently unprofitable per unit of revenue; they are minor contributors purely in scale. This indicates the business's aggregate financial performance is closely tied to Chocolate-specific demand and cost trends.

B. Product-Level Profit Concentration (Pareto)

Ranking all fifteen products by gross profit reveals that five products (33% of the catalog) account for approximately 80% of both total gross profit and total sales — a notably tight alignment suggesting these top products are consistently high-margin rather than high-volume/low-margin outliers. The remaining ten products contribute the residual 20% of both metrics.

C. Cost and Pricing Diagnostics

Applying a joint screening rule (gross margin below 35% and cost-to-sales ratio above 50%), only one product across the entire fifteen-item catalog was flagged: Kazookles, with a gross margin of 7.69% and a cost-to-sales ratio of 92.31%, meaning approximately 92 cents of cost is incurred per dollar of revenue generated. All other fourteen products returned a healthy status under this rule, most between 46% and 80% gross margin, indicating strong pricing discipline across the

catalog overall with Kazookles as an isolated, clear exception.

D. Geographic Concentration

Revenue is concentrated in a small number of states, as shown in Table III. The top five states account for approximately 51% of total company revenue.

TABLE III

TOP FIVE STATES BY REVENUE CONTRIBUTION

State	Sales	Contrib. %
California	$27,917	20%
New York	$15,541	11%
Texas	$13,416	9%
Pennsylvania	$8,027	6%
Washington	$6,921	5%

Collectively, these findings indicate the business, as recorded, is concentrated rather than diversified — by division, by product, and by geography — a consistent pattern across all four analytical dimensions examined.

V. DASHBOARD IMPLEMENTATION

The findings above were operationalized into a five-page interactive Power BI dashboard, summarized in Table IV and illustrated in Figs. 1–6.

TABLE IV

DASHBOARD PAGE SUMMARY

Page	Primary Function
Overview	KPIs, filters, narrative
Product Profitability	Margin leaderboard, scatter
Division Performance	Revenue vs. profit, map
Cost Diagnostics	Threshold repricing flags
Profit Concentration	Pareto, geo. table



Fig. 1. Overview page — headline KPIs, global filters, and dynamic narrative insight.

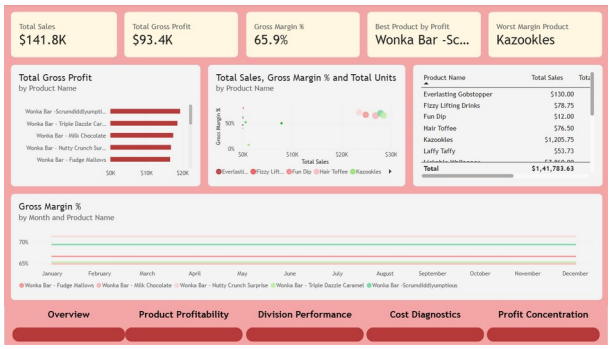


Fig. 2. Product Profitability page — margin leaderboard, cost/margin scatter, product detail table.

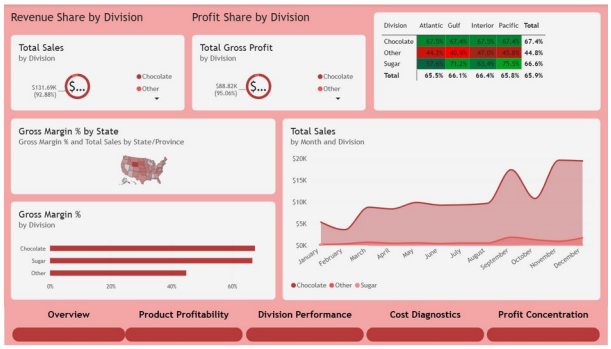


Fig. 3. Division Performance page — revenue vs. profit share, state margin map, region matrix.

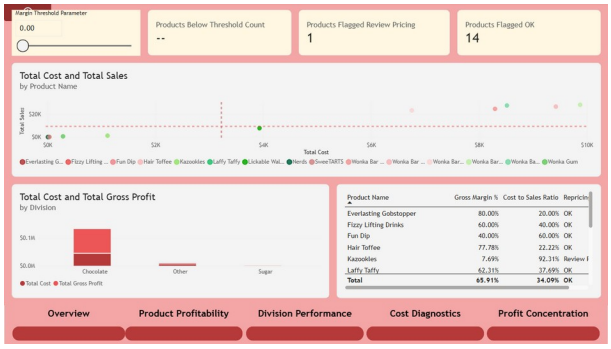


Fig. 4. Cost Diagnostics page — adjustable margin-threshold slider with dynamic repricing flags.

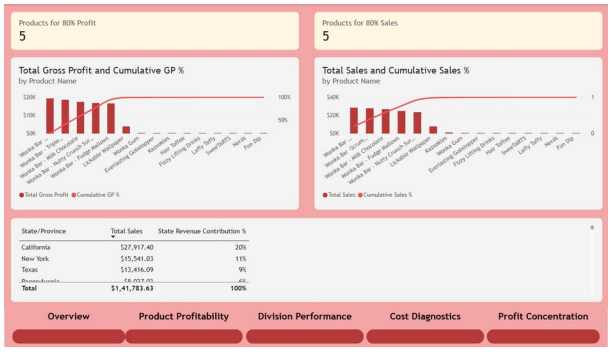


Fig. 5. Profit Concentration page — Pareto analysis for profit and revenue, geographic concentration table.

A persistent five-button navigation bar (Overview, Product Profitability, Division Performance, Cost Diagnostics, Profit

Concentration) appears at the foot of every analytical page, allowing a reader to move between pages, or return to the Overview page, from anywhere in the report. A sixth, drill-through page (Fig. 6) is reachable from any product-level visual and provides a single-product monthly sales trend, filtered automatically to the selected product via Power BI's native drillthrough mechanism, with a Back button to return to the originating page.

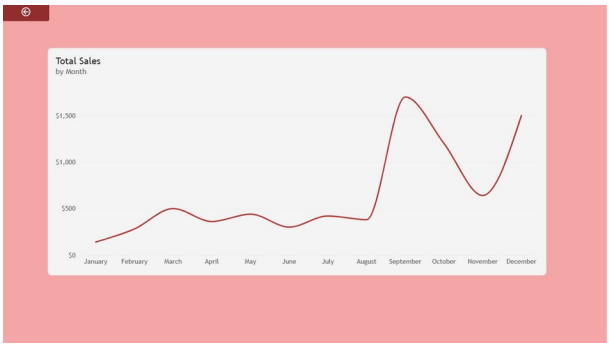


Fig. 6. Product Deep Dive — drill-through page showing single-product monthly sales trend.

The Date, Division, and Region filters are synchronized from the Overview page across the report using Power BI's Sync Slicers feature, while remaining visually present only on the Overview page to preserve a clean, uncluttered layout on the four analytical pages; a Product Name search slicer is also provided on the Overview page. The Cost Diagnostics page additionally implements a What-If numeric parameter bound to a slicer control, allowing a user to interactively adjust the margin threshold and observe the resulting change in flagged at-risk products in real time — directly satisfying the interactivity requirement of the original project brief.

VI. RECOMMENDATIONS

A. Portfolio and Diversification

- Treat Chocolate's dominance as a concentration risk rather than solely a success indicator; evaluate whether Sugar and Other divisions are underinvested relative to their demonstrated margin health.
- Prioritize supply-chain stability and consistent pricing for the five products driving approximately 80% of profit, as these represent the business's most valuable assets on a per-unit basis.

B. Pricing and Cost Action

- Conduct an immediate cost and pricing review of Kazookles; at a 92.31% cost-to-sales ratio this product is likely operating near breakeven once indirect costs are considered.
- Re-run the repricing screen periodically using the dashboard's adjustable threshold rather than treating the current 35% cutoff as fixed.

C. Geographic Strategy

- Investigate whether concentration in the top five states reflects deliberate strategy or unmanaged organic growth, and consider targeted expansion into underrepresented states to reduce single-region dependency.

D. Data Governance

- Resolve the Ship Date data quality issue at the source system before undertaking any shipping-efficiency or logistics analysis on this dataset.

VII. CONCLUSION

This work demonstrates that an aggregate gross margin of 65.9% for Nassau Candy Distributor conceals substantial structural concentration across division, product, and geography. The interactive Power BI dashboard delivered alongside this paper equips stakeholders to monitor these dynamics on an ongoing basis, with the single clearest actionable finding being the urgent need to review the pricing of Kazookles. Future work should incorporate time-series cost data and external demand signals to further disambiguate pricing-error from cost-inflation causes of margin erosion.

ACKNOWLEDGMENT

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